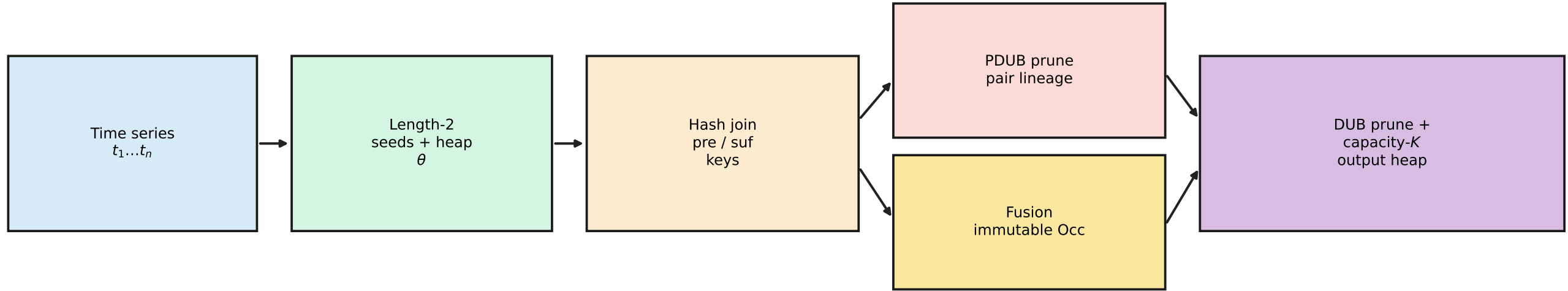


TK-HJ-OPF: exact top- K OPF under exponential forgetting

$$w_j = e^{-k(n-j)} \cdot \text{no minsup} \cdot \text{strict rank encoding}$$



Safe because $w_{j+d} = e^{kd} w_j$. PUB (direct child) cannot prune a lineage.

Locked Java campaign (2026-09-06): geomean 19.63× vs exhaustive hash-join on OPF DB1-DB8.